

HCAHPS Patient Survey

1) Business Task:

Analyze HCAHPS data to decide whether hospitals made improvements in their quality of care over the 9 years studied, areas of opportunity for growth, and recommendations that the hospitals can utilize to improve patient experiences.

2) Data Description:

This is the Hospital Consumer Assessment of Healthcare Providers and Systems (HCAHPS) data which has been provided by Maven Analytics.

3) Data Cleaning:

- Checked for missing values across all tables
- Removed empty rows in responses, state_responses, and states tables
- Validated percentage columns, ensuring no values exceeded 100%

4) Analysis Summary:

Hospitals have not shown much improvement in the quality of care during the period analyzed. The analysis includes 10 patient experience measures:

1. Communication with Nurses
2. Communication with Doctors
3. Responsiveness of Hospital Staff
4. Communication about Medicines
5. Discharge Information
6. Care Transition
7. Cleanliness of Hospital Environment
8. Quietness of Hospital Environment
9. Overall Hospital Rating
10. Willingness to Recommend the Hospital

The HCAHPS survey response categories are based on the level of positive experiences. Top-box responses are where patients report having the most consistent positive experiences where they would answer with “always” or “strongly agree.” Middle-box and are less consistent positive experiences with answers like “usually” or “agree” and Bottom-box responses are the least positive experiences with answers like “sometimes or never” or “strongly disagree.”

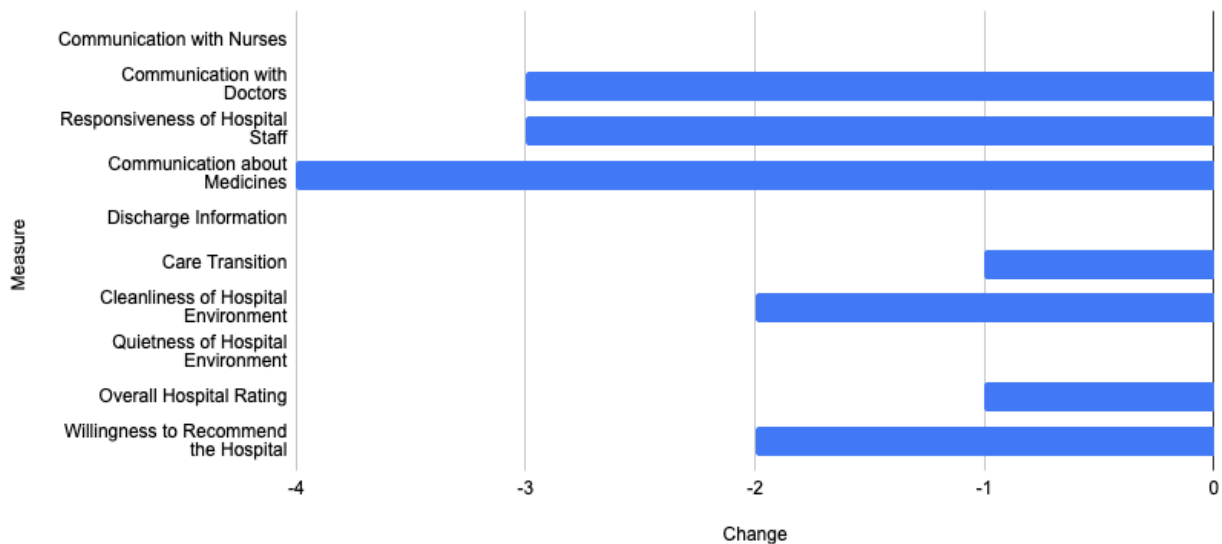
Of these 10 measures, 7 decreased while 3 remained unchanged between the 2015 and 2023 patient survey reports. No measure showed an increase in the top-box

responses. Of the 7 measures that decreased, the largest decrease was in Communication about Medicines by 4 percentage points with the average decrease among the top-box responses being by 2.28 percentage points. The second largest decline was a tie between Communication with Doctors and Responsiveness of Hospital Staff, both decreasing by 3 percentage points.

No specific areas improved but some areas declined less than others and some were stable based on top-box responses. The measures that were stable or had no change were Communication with Nurses, Discharge Information, and Quietness of Hospital Environment. Among the stable measures, the overall responses still shifted towards the middle and bottom box-responses which are less-positive responses. A major area of opportunity is Communication about Medicine. It had the largest decline of top-box responses and has the most bottom-box responses of the 10 measures analyzed.

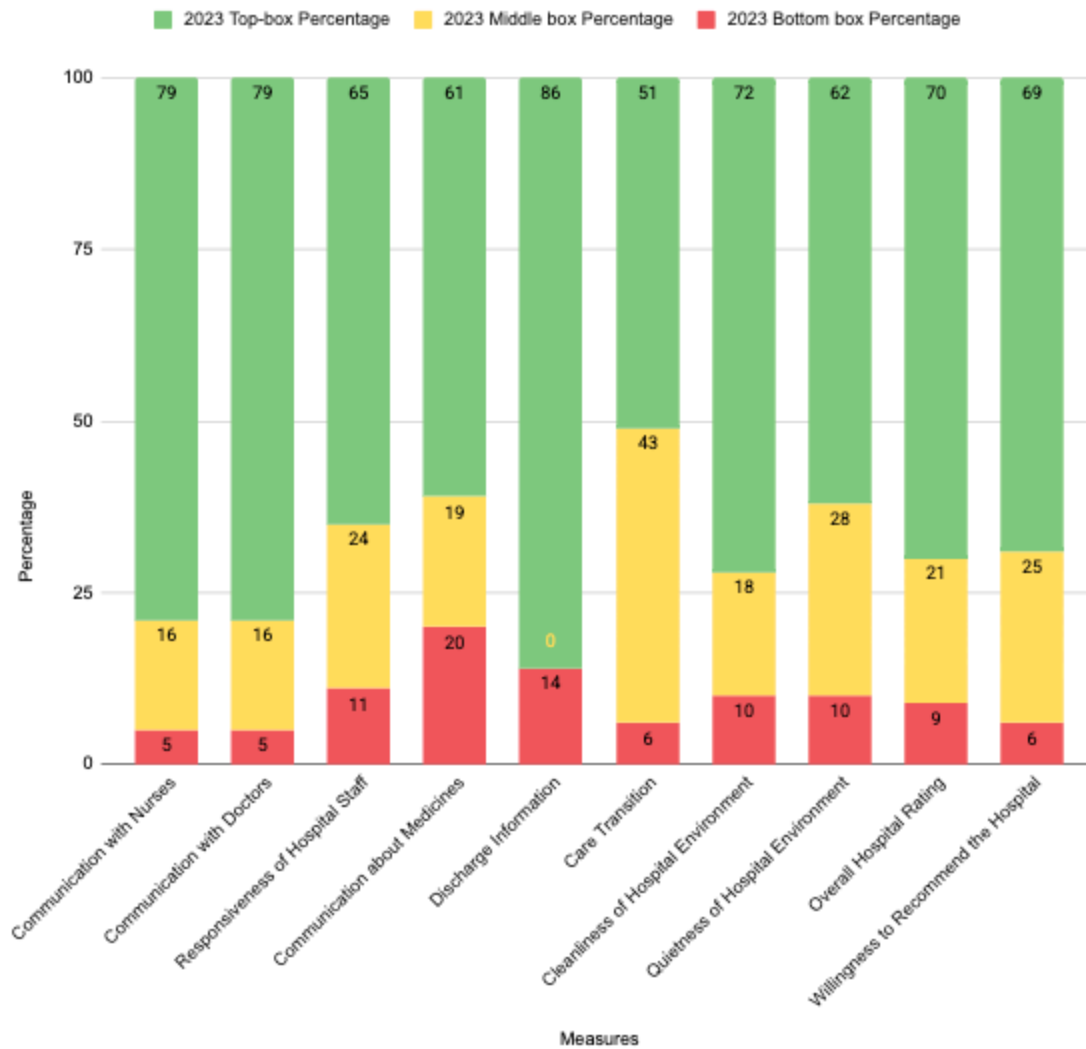
5) Visualizations:

Top-Box Change 2015 - 2023



Change in top-box responses from 2015-2023 by Measures analyzed.

2023 Patient Experience Responses by Measure



Distribution of responses across top-box, middle-box, and bottom-box answers to 2023 Patient surveys.

6) Recommendations:

- Improve communication about medications to patients
- Focus on improving areas with higher bottom-box responses